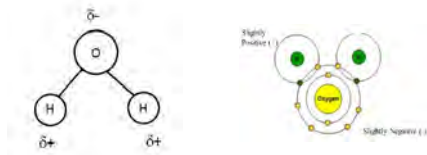


| Question number | Answer                                                                                                                                                                                       | Additional guidance                                                                                                                          | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(a)            | <p>A diagram that includes the following:</p> <ul style="list-style-type: none"> <li>2 hydrogens joined to an oxygen (by covalent bonds) (1)</li> <li>the charge distribution (1)</li> </ul> |  <p>e.g. lines, overlapping circles, shared electrons</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                           | Additional guidance                                                                                                                                                                                                             | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)(i)         | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>for F increase in temperature increases solubility (1)</li> <li>for G increase in temperature increases up to 30°C and then decreases the solubility (1)</li> <li>for H temperature has no effect (on solubility) (1)</li> </ul> | <p><b>ACCEPT</b> 'It' for solubility throughout</p> <p><b>ACCEPT</b> positive correlation</p> <p>Do not piece together</p> <p><b>ACCEPT</b> solubility {remains constant / does not change} with an increase in temperature</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                                               | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)(ii)        | <p>The only correct answer is B 1.67</p> <p><i>A is incorrect because <math>50 \div 30 = 1.67</math></i></p> <p><i>C is incorrect because <math>50 \div 30 = 1.67</math></i></p> <p><i>D is incorrect because <math>50 \div 30 = 1.67</math></i></p> | (1)  |